

Further Development of Mathematical Model for Bluefin

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Abstract

The mathematical model of the ship is the core of the ship motion simulation and control problem. This paper mainly studies the mathematical modelling and simulation of Bluefin using low-cost sensors. This paper focuses on the study of the ship's free running turning circle motion, using the MMG model. By analysing the hull force, propeller force and rudder force and assuming that the environmental factors are no wind and no current, the relevant parameters can be calculated to build a mathematical model of the ship. Based on this model, MATLAB is used for simulation.

Keywords

Ship Motion, Mathematical model, MATLAB simulation, MMG model, Bluefin, Low cost, Turning circle trajectory

Nomenclature

Notation	Description	Nominal Unit
M	Mass of ship	kg
L _{oa}	Length over all	m
L _{BP}	Length between perpendicular	m
B	Breadth	m
d	Mean draft	m
I	Moment of inertia	kgm ²
ρ	Density of water	kg/m ³
V	Displacement	m ³
C _b	Block coefficient	-
t _R	Resistance deduction coefficient	-
L _R	Rudder length	m
W _R	Rudder width	m
S	Rudder area	m ²
∂_H	Mutually intervened coefficient between ship hull and rudder	-
λ	The aspect ratio of the rudder	-
f _a	Slope due to the inclined angle of single rudder normal pressure	-
X _H	longitudinal distance from the centre of the ship's lateral force to the centre of gravity	m
k	Experimental coefficient	-
n	Propeller Speed	rev/s
w	Wake coefficient of single propeller	-

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γ_R	Rectification coefficient	-
l_R	Rudder vertical coordinate	m
m_x	Addition mass	kgs ² /m
m_y	Addition mass	kgs ² /m
I_{ZZ}	Addition moment of inertia	kgs ² /m
J_{ZZ}	Addition moment of inertia	kgs ² /m
$I_{ZZ}+J_{ZZ}$	Z-axial inertial moment & added one	kgs ² /m
t_{po}	Thrust deduction coefficient	-
μ	Coefficient of viscosity (freshwater)	-
k	Test coefficient	-
C_A	Ship model coefficient	-
∇	Drainage volume	m ³
S_w	Hull wet area	m ²
K_T	Thrust coefficient	-
J	Propulsion coefficient	-
A_R	Area of the rudder	m ²
l_R	Longitudinal coordinate of rudder	m
g	Acceleration of gravity	m/s ²
X'_V	Derivative of hydrodynamic force to surge	-
X'_{Vr}	Derivative of hydrodynamic force to surge	-
X'_r	Derivative of hydrodynamic force to surge	-
X'_{vv}	Derivative of hydrodynamic force to surge	-
X'_{rr}	Derivative of hydrodynamic force to surge	-
Y'_v	Derivative of hydrodynamic force to sway	-
Y'_{vr}	Derivative of hydrodynamic force to sway	-
Y'_r	Derivative of hydrodynamic force to sway	-
Y'_{vv}	Derivative of hydrodynamic force to sway	-
Y'_{rr}	Derivative of hydrodynamic force to sway	-
N'_v	Derivative of hydrodynamic force to yaw	-
N'_{vr}	Derivative of hydrodynamic force to yaw	-
N'_r	Derivative of hydrodynamic force to yaw	-
N'_{vv}	Derivative of hydrodynamic force to yaw	-
N'_{rr}	Derivative of hydrodynamic force to yaw	-

1 Introduction

With the continuous improvement of automatic control requirements such as autopilot, automatic berthing, and dynamic positioning, simulation by deducing and researching mathematical models can help surface ships operate more accurately and safely. The mathematical model is a kind of simulation. It uses mathematical symbols, mathematical formulas, programs, graphics to explain some objective phenomena, or to predict the development of future phenomena, or to provide some optimal strategies to control the development of a phenomenon. (ANSARI,2016) The establishment of mathematical model needs people to use all kinds of mathematical knowledge to observe and analyse practical problems.

Developing a new MMG mathematic model for Bluefin using the free-running test. The free-running test is a ship model test driven by ship model propeller, which can predict the speed of the ship and analyse the interaction between propeller and hull. It can judge the comprehensive performance of ship, rudder, and propeller. This method is the best way to study the interaction between the ship hull and propeller. Through the free-running model test, we can know that the ship's

information includes estimating the ship's navigation performance (the relationship between ship power and ship speed) and analysing the propeller efficiency.

This project has been studied before, but the former researchers did not get a correct mathematical model and did not simulate an accurate trajectory through MATLAB. The statement of the modelling problem for previous research work at AMC are presented below:

Jiang (2016) focuses on modelling and simulating Bluefin models with low-cost sensors. In this paper, the first order Nomoto mathematical model of the Bluefin is presented. The least square method and recursive least square method are proposed to estimate the hydrodynamic coefficients. In the free running model test, the Bluefin's propeller broke. Jiang suggested replacing the GPS sensor to obtain more accurate position information.

Hao (2017) focused on the development of MMG mathematical model for bluefin using free running tests. Hao didn't complete the Bluefin mathematical model. The simulation result for her model is not correct. During the experiment, because the sensors carried by ship and the laptop control panel are too far apart, the sensors cannot obtain signals which leads to the propulsion system out of control.

Liang (2018) focuses on the berthing and unberthing of the surface ship model. The simulation based on PID and FLT is designed on Shioji Maru. Liang's propulsion system model is not correct, and there are many uncertainties in the model. In the actual experiment, there is no GPS data for turning verification. In his report, Liang mentioned two major problems, namely, system hardware problems and control program failure to generate error signals for the system.

Since previous researchers did not build an accurate MMG mathematical model for bluefin, the scope of the project is to review the current mathematical model, analyse its uncertainty, modify the shortcomings, and establish a new mathematical model to test. The skills involved in this project include understanding and applying ship dynamics and hydrodynamics and using MATLAB simulation. This project is to establish a free-running model test. This test method can directly determine the manoeuvrability of the ship. The mathematical model of ship manoeuvrability is very important in simulation. This project is to continue the previous experimental project. In this project, Bluefin's ship model is simulated, and the simulation result proves the new MMG mathematical model is suitable.

The first and second parts of the second section of this paper respectively introduce the ship's motion coordinate system and explain the previous research results. In the third part, MMG mathematical model is defined and established. The third section uses MATLAB to simulate and verify the MMG model. Finally, the fourth section is the conclusion and suggestions for further study.

2 The Establishment of Ship Mathematical Model

The MMG model is a mathematical model of linear ship motion simplified by the characteristics of ship motion. MMG's separation modeling thought separately considers the hull, propeller and rudder.

The scaled model used in the experiment is Bluefin and it is a scale model of the Australia Maritime College training vessel FTV Bluefin. The vessel is used to train both engineering and seafaring students at AMC.

2.1 Bluefin Dimensions

Table 1. Dimensions of Bluefin Model Scale

m	Mass	64.550	kg
Loa	Length Over All	1.725	m
Lpp	Length Between Perpendicular	1.569	m
B	Breadth	0.500	m
d	Draft	0.193	m
I	Inertia Moment	10.450	kg

Table 2. Specification of The Motors

Item	Power supply
DC motor for propeller	12 V
DC motor for thruster	6 V
Servo motor	12 V

2.2 Bluefin Motion Coordinate System

To describe the motion of a ship, the coordinate system must be determined first. Here, a fixed coordinate system X, Y and Z consistent with the surface of the earth and a ship motion coordinate X_o, Y_o and Z_o are used. The fixed coordinate system X, Y and Z uses any point as the origin to establish a three-dimensional coordinate system. The ship motion coordinate system takes the ship's center of gravity as the coordinate origin, the ship direction as the positive x axis, the starboard side as the positive y axis, and the cross-section perpendicular to the center of gravity as the z axis.

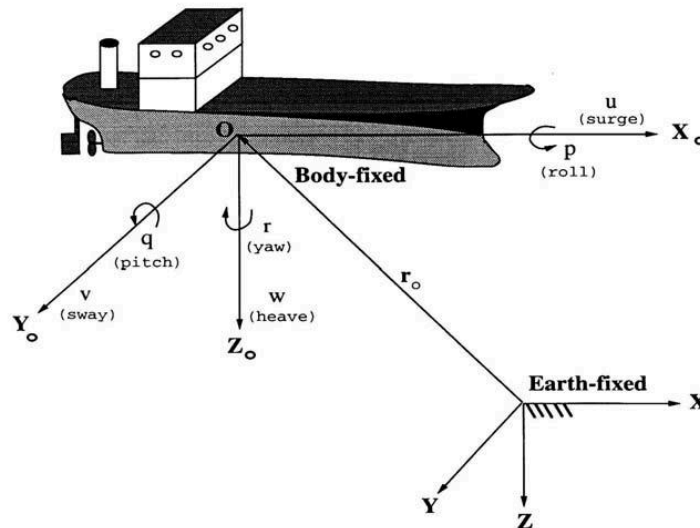


Figure 1: Ship motion coordinate system (Fossen, 1994)

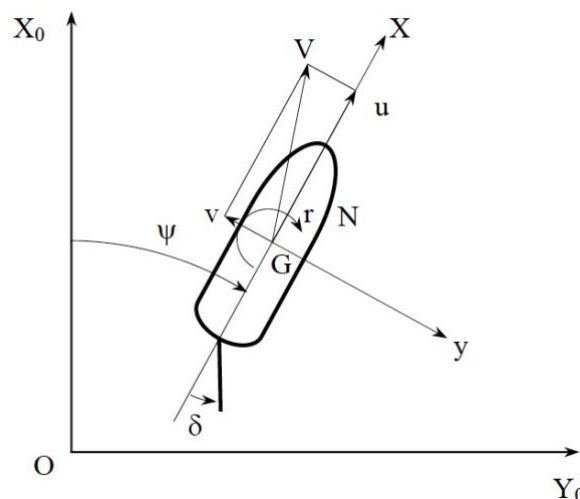


Figure 2: Ship's coordinate in horizontal plane (Nguyen, 2014)

2.3 Review and Assessment on the Models Developed Previously

Kinetic Equations

$$X = mvr - \frac{1}{2}\rho S_w C_t |u|u + \frac{1}{2}\rho L_{pp} dV^2 (X_v |v'| + X'_{vr} v' r' + X'_r |r'| + X'_{vv} v' v' + X'_{rr} r' r') + (1 - t_p) \rho n^2 D_p^4 - (1 - t_R) F_N \sin \delta \quad (1)$$

$$Y = -mvr + \frac{1}{2}\rho L_{pp} dV^2 (Y_v v' + Y'_{vr} v' |r'| + Y'_r r' + Y'_{vv} v' |v'| + Y'_{rr} r' |r'|) + \frac{T_b S_b}{10} - (1 - t_R) F_N \sin \delta \quad (2)$$

$$N = \frac{1}{2}\rho L_{pp} dV^2 (N_v v' + N'_{vr} v' |r'| + N'_r r' + N'_{vv} v' |v'| + N'_{rr} r' |r'|) + \frac{T_b X_b S_b}{10} - (x_R + a_H x_H) F_N \cos \delta \quad (3)$$

m: Mass of the vessel (kg s²/m)

I_z: Moments of inertia of yaw (kg m s²)

u and v: Velocity of surge and sway (m/s)

r: Yaw angular velocity (rad/s)

L_{pp}: Length between perpendiculars (m)

d: Draft (m)

S_w: Immersed area of ship hull (m²)

C_t: Hull resistant coefficient

v' : Non-dimensional velocity in x -direction ($v'=v/V$)

r' : Non-dimensional yaw angular velocity ($r'=rL_{pp}/V$)

X_v, X_r, X_{vv}, X_{rr}, X_{vr}: Non-dimensional derivative coefficients

Y_v, Y_r, Y_{vv}, Y_{rr}, Y_{vr}: Non-dimensional derivative coefficients

N_v, N_r, N_{vv}, N_{rr}, N_{vr}: Non-dimensional derivative coefficients

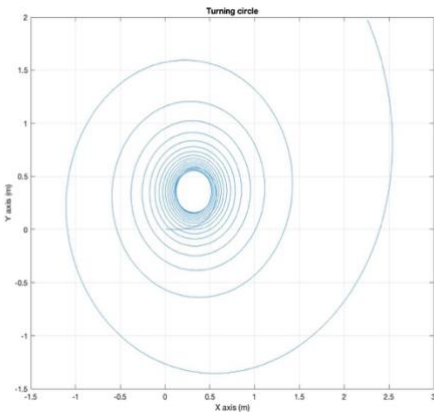


Figure 3: Hao (2018) Simulated turning circle trajectory with 10 degree

The ship should starts move along y axis first and then rotate clockwise.

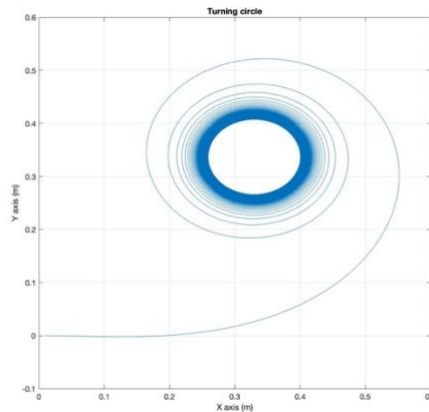


Figure 4: Hao (2018) Simulated turning circle trajectory with -10 degree

The ship should starts move along y axis first and then rotate counter clockwise.

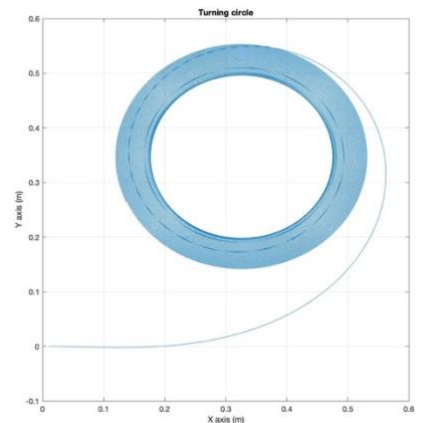


Figure 5: Hao (2018) Simulated turning circle trajectory with 0 degree

The ship should move along y axis and the trajectory should be a vertical line.

2.4 Further Development

MMG Model

$$m * (\dot{u} - vr) = x = X_H + X_P + X_R + X_T \quad (4)$$

$$m * (\dot{v} + ur) = y = Y_H + Y_P + Y_R + Y_T \quad (5)$$

$$I_{ZZ} * \dot{r} = N - Y * X_C = N_H + N_P + N_R + N_T \quad (6)$$

During test, fluid moment relative to the centre of the ship, so it is necessary to modify the moment to the relative centre of the ship, X_c is the coordinate value of the x-axis of the ship's centre in the coordinate system.

$$(m + m_x) * \dot{u} - (m + m_y) * vr = x = X_H + X_P + X_R + X_T \quad (7)$$

$$(m + m_y) * \dot{v} + (m + m_x) * ru = y = Y_H + Y_P + Y_R + Y_T \quad (8)$$

$$(I_{ZZ} + J_{ZZ}) * \dot{r} = N - Y * X_C = N_H + N_P + N_R + N_T \quad (9)$$

Where

$$f_o = \tau_1 = [X, Y, Z]^T \text{-----External forces}$$

$$m_o = \tau_2 = [K, M, N]^T \text{-----Moment of external force about ship's centre of gravity}$$

$$v_o = v_1 = [u, v, w]^T \text{-----Linear velocity of } X_o, Y_o, Z_o$$

$$\omega = v_2 = [p, q, r]^T \text{-----Angular velocity of } X_o, Y_o, Z_o$$

$$r_G = [X_G, Y_G, Z_G]^T \text{-----Centre of gravity}$$

It is assumed that the origin of the coordinate system coincides with the centre of the ship and centre of the gravity, and the inertia forces and moments are converted into the additional mass of the ship along the x and y axis and additional mass moment of inertia around the z axis.

$$X_H = X_{NL} \quad (10)$$

$$Y_H = Y_L + Y_{NL} \quad (11)$$

$$N_H = N_L + N_{NL} \quad (12)$$

$$X_H = X(u) + X_{vv}V^2 + X_{vr}vr + X_{rr}r^2 \quad (13)$$

$$Y_H = Y_vv + Y_rr + Y_{NL} \quad (14)$$

$$N_H = N_vv + N_rr + N_{NL} \quad (15)$$

Y_{NL} , N_{NL} is nonlinear fluid dynamics and moment

$$Y_{NL}, N_{NL} = -\frac{1}{2} * \rho * d \int_{-\frac{1}{2}}^{\frac{1}{2}} (v + xr) * |v + xr| C_D(x) dx \quad (16)$$

It is a high order small amount relative to linear fluid dynamics and can be omitted.

$$X(u) = \frac{1}{2} * \rho * v^2 * L * d * X'(u') \quad (17)$$

$$X'(u') = X'_{uu} * u'^2 \quad (18)$$

$$X'_{uu} = \frac{S}{L * d} * C_t \quad (19)$$

C_t is the total drag coefficient of the ship's direct navigation

Hull

$$X_H = \frac{1}{2} \rho S_w C_t |u|u + \frac{1}{2} \rho L_{pp} dV^2 (X_v |v'| + X'_{vr} v' r' + X'_r |r'| + X'_{vv} v' v' + X'_{rr} r' r') \quad (20)$$

$$Y_H = \frac{1}{2} \rho L_{pp} dV^2 (Y_v v' + Y'_{vr} v' |r'| + Y'_r r' + Y'_{vv} v' |v'| + Y'_{rr} r' |r'|) \quad (21)$$

$$N_H = \frac{1}{2} \rho L_{pp} dV^2 (N_v v' + N'_{vr} v' |r'| + N'_r r' + N'_{vv} v' |v'| + N'_{rr} r' |r'|) \quad (22)$$

m: Mass of the vessel ($\text{kg s}^2/\text{m}$)

I_z: Moments of inertia of yaw (kg m s^2)

u and v: Velocity of surge and sway (m/s)

r: Yaw angular velocity (rad/s)

L_{pp}: Length between perpendiculars (m)

d: Draft (m)

S_w: Immersed area of ship hull (m^2)

C_t: Hull resistant coefficient

v' : Non-dimensional velocity in x -direction ($v'=v/V$)

r' : Non-dimensional yaw angular velocity ($r'=rL_{pp}/V$)

Rudder

$$X_R = (1 - t_R) * F_N \sin \delta \quad (23)$$

$$Y_R = (1 + \partial_H) * F_N \cos \delta \quad (24)$$

$$N_R = (\partial_R + \partial_H X_H) * F_N \cos \delta \quad (25)$$

$$K_R = -Y_R Z_R \quad (26)$$

X_R , Z_R is rudder force centre coordinates

F_N is rudder normal force

$$F_N = -\frac{1}{2} * \rho * A_R * f_\alpha * u_R^2 \sin \alpha_R \quad (27)$$

A_R is area of rudder

f_α is the slope due to inclined angle of single plate rudder normal pressure

U_R is effective inflow velocity due to rudder

α_R is effective inflow angle due to rudder

∂_H is mutually intervened coefficient between ship hull and rudder

$$\partial_H = 0.6784 - 1.3374 * C_b + 1.8891 * C_b^2 \quad (28)$$

C_b is the block coefficient

$$C_b = \frac{V}{L * B * d} \quad (29)$$

L is water line length

B is width of ship

d is Draught

V is the displacement

$$V = \frac{M}{\rho} \quad (30)$$

∂_R is inflow angle due to rudder

$$\partial_R = \delta + \arctan \frac{v_R}{u_R} \quad (31)$$

$\frac{v_R}{u_R}$ is very small, so $\partial_R = \delta$

$$v_R = \gamma_R * (v + l_R r) \quad (32)$$

$$\partial_R = \delta + \frac{\gamma_R}{u_R} * (v + l_R r) \quad (33)$$

$$F_N = -\frac{1}{2} * \rho * A_R * f_\alpha * u_R^2 \sin[\delta + \frac{\gamma_R}{u_R} * (v + l_R r)] \quad (34)$$

f_α is slope due to the inclined angle of single rudder normal pressure (Prandtl equation)

$$f_\alpha = \frac{2\pi\lambda}{2 + \lambda} \quad (35)$$

λ is the aspect ratio of the tablet

$$\lambda = \frac{L_R^2}{S} \quad (\lambda = 0.5 \sim 3.0) \quad (36)$$

S is rudder's area

X_H is the longitudinal distance from the centre of the ship's lateral force to the centre of gravity of the ship induced by steering which about (0.4-0.5) L.

$$X_H = -L * (0.4 + 0.1 * C_b) \quad (37)$$

u_p is the inflow velocity at the propeller

$$u_p = (1 - w) * u \quad (38)$$

w is the wake coefficient of single propeller

$$w = 0.5 * C_b - 0.05 \quad (39)$$

u_R is the effective longitudinal velocity into the rudder (Fujii equation)

$$u_R = u_p \sqrt{1 + k s^{1.5}} \quad (40)$$

$ks^{1.5}$ is the growth rate of the propeller

k is the experimental coefficient

Taking a pitch ratio of 0.0717, then $P = p * D = 0.00717$

s is the slippage coefficient of the propeller

$$s = 1 - \frac{u_p}{n * P} \quad (41)$$

n is the propeller speed

$$u_R = 0.836935 * u \sqrt{1 + 6.3 \left(1 - \frac{(0.856113 * u)}{0.00717n} \right)^{1.5}} \quad (42)$$

γ_R is rectification coefficient

n is Propeller speed

$$\gamma_R = 1.1633 - 1.9828 * C_b + 1.3902 * C_b^2 \quad (43)$$

l_R is the rudder vertical coordinate

$$l_R \approx -0.9 * L \quad (44)$$

$$X_R = -(1 - 0.449821) * F_N * \sin(\delta) \quad (45)$$

$$Y_R = -(1 + 0.443853) * F_N * \cos(\delta) \quad (46)$$

$$N_R = -(0.646875 + 0.443853 * 0.7569) * F_N * \cos(\delta) \quad (47)$$

m_x, m_y is addition mass. (according to Ming's equation)

$$\frac{m_x}{m} = \frac{1}{100} [0.398 + 11.97 * C_b * \left(1 + 3.73 * \frac{d}{B} \right) - 2.89 * C_b * \frac{L}{B} * \left(1 + 1.13 * \frac{d}{B} \right) + 0.1758 * C_b * \left(\frac{L}{B} \right)^2 * \left(1 + 0.541 * \frac{d}{B} \right) - 1.107 * \frac{L}{B} * \frac{d}{B}] \quad (48)$$

$$\frac{m_y}{m} = 0.882 - 0.54 * C_b * \left(1 - 1.6 * \frac{d}{B} \right) - 0.156 * \frac{L}{B} * (1 - 0.673 * C_b) + 0.826 * \frac{d}{B} * \frac{L}{B} * \left(1 - 0.678 * \frac{d}{B} \right) - 0.638 * C_b * \frac{d}{B} * \frac{L}{B} * (1 - 0.669 * \frac{d}{B}) \quad (49)$$

I_{ZZ}, J_{ZZ} is addition moment of inertia (according to Ming's equation)

$$I_{ZZ} = 0.05 * \rho * V * L^2 \quad (50)$$

$$\frac{J_{ZZ}}{L^2} = \frac{1}{100} * [33 - 76.85 * C_b * (1 - 0.784 * C_b) + 3.43 * \frac{L}{B} * (1 - 0.63 * C_b)] \quad (51)$$

Shallow water formula

When the ratio of water depth to draft is less than 3, the ship is considered to be sailing in shallow water, and the force of inertial current increases with the decrease of water depth.

$$\frac{m_x}{m} = \left[\left(\frac{H}{d} - 1 \right)^{1.3} + 3.7695 + 1.142 * \frac{B}{d} - 0.2329 * \frac{L}{d} - 3.4274 * C_b \right] / \left(\frac{H}{d} - 1 \right)^{1.3} \quad (52)$$

$$\frac{m_y}{m} = \left[\left(\frac{H}{d} - 1 \right)^{0.82} + 0.4127 + 0.0320 * \frac{B}{d} + 0.01293 * \left(\frac{B}{d} \right)^2 \right] / \left(\frac{H}{d} - 1 \right)^{0.82} \quad (53)$$

$$\frac{J_{ZZ}}{m} = \left[\left(\frac{H}{d} - 1 \right)^{0.82} + 0.4127 + 0.01922 * \frac{B}{d} + 0.00554 * \left(\frac{B}{d} \right)^2 \right] / \left(\frac{H}{d} - 1 \right)^{0.82} \quad (54)$$

Propeller

$$X_p = (1 - t_{po}) * \rho * n^2 * D_e^4 * K_T * J \quad (55)$$

$$Y_p = 0 \quad (56)$$

$$N_p = 0 \quad (57)$$

t_{po} is thruster deduction coefficient

K_T is thrust coefficient

J is advance ratio

$$J = \frac{v_a}{n * D} \quad (58)$$

v_a is advance velocity

According to Cornwall Model Boats pitch value is 71.7 mm

$$\frac{P}{D_p} = 0.717 \quad (59)$$

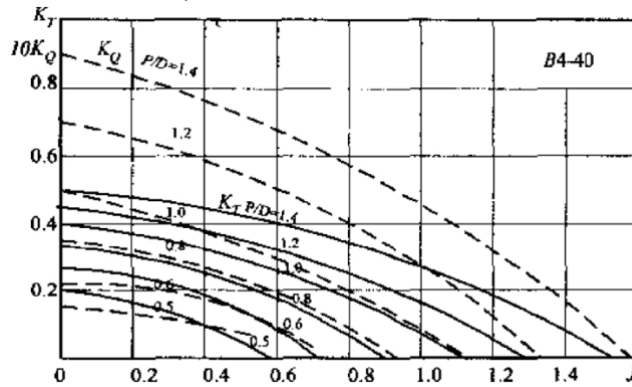


Figure 6: K_T Vs J (Yang,1996)

K_T can be find in figure 3, which is 0.25.

The speed of the ship relative to the riverbank during counter current.

$$v_1 = v_{sw} - v_{rb} \quad (60)$$

v_{sw} is speed in still water

v_{rb} is speed of water flow relative to the riverbank

The speed of movement of the ship and water is opposite, so the speed of the ship relative to the water shows below:

$$v_a = v_1 + v_{rb} = v_{sw} - v_{rb} + v_{rb} = v_{sw} \quad (61)$$

$$X_p = 4.0229 * K_T * v_a \quad (62)$$

w_p is effective wake coefficient at propeller position (according to Holtflow's formula)

$$w_p = 0.3095 * C_b + 10 * C_v * C_b - 0.23 * \frac{D_p}{\sqrt{B * d_m}} \quad (63)$$

D_p is propeller diameter

C_v is coefficient of viscous resistance

$$C_v = (1 + k) * C_F + C_A \quad (64)$$

K is test coefficient

C_F is friction drag coefficient (according to 1957 ITTC)

$$C_f = \frac{0.075}{(\lg(R_e) - 2)^2} \quad (65)$$

R_e is Reynolds number

$$R_e = \frac{L * v}{\mu} \quad (66)$$

μ is coefficient of viscosity

C_A is ship model coefficient

$$C_A = 0.006 * (L + 100)^{-0.16} - 0.00205 + 0.003 * \sqrt{\frac{L}{7.5 * C_b^4}} * (0.04 - C_4) \quad (67)$$

$$C_4 \begin{cases} \frac{d_F}{L} & \frac{d_F}{L} \leq 0.04 \\ 0.04 & \frac{d_F}{L} > 0.04 \end{cases} \quad (68)$$

S_w is hull wet area

$$S_w = L * B * d * \left(\frac{1}{d} + \frac{2}{B} + \frac{2}{L} \right) - 4.4 \left[\nabla * \left(\frac{1}{C_b} - 1 \right) \right]^{\frac{2}{3}} \quad (69)$$

∇ is drainage volume

When the block coefficient decrease, the drainage volume and hull wet area also decrease.

$$L * B * d = \frac{\nabla}{C_b} \quad (70)$$

3 Simulation Study for Ship Manoeuvring

According to the standard manoeuvring test (Fossen, 1994), MATLAB simulation will be carried out with Straight Line Test and Turning Circle Test. The purpose of the turning test is to determine the speed of horizontal turning, especially the speed of large rudder turning and the size of the required water area, so as to determine the turning capacity of the ship. The rotation circle test process is to turn the rudder to a certain angle until the test circle reaches 540 degrees. It can show ship manoeuvrability under strong and weak manoeuvres.

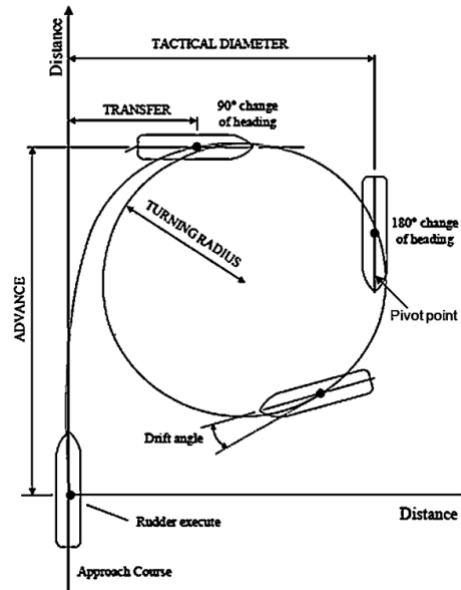


Figure 7: Turning circle trajectory (Shipknow, 2018)

3.1 MATLAB Simulation Result

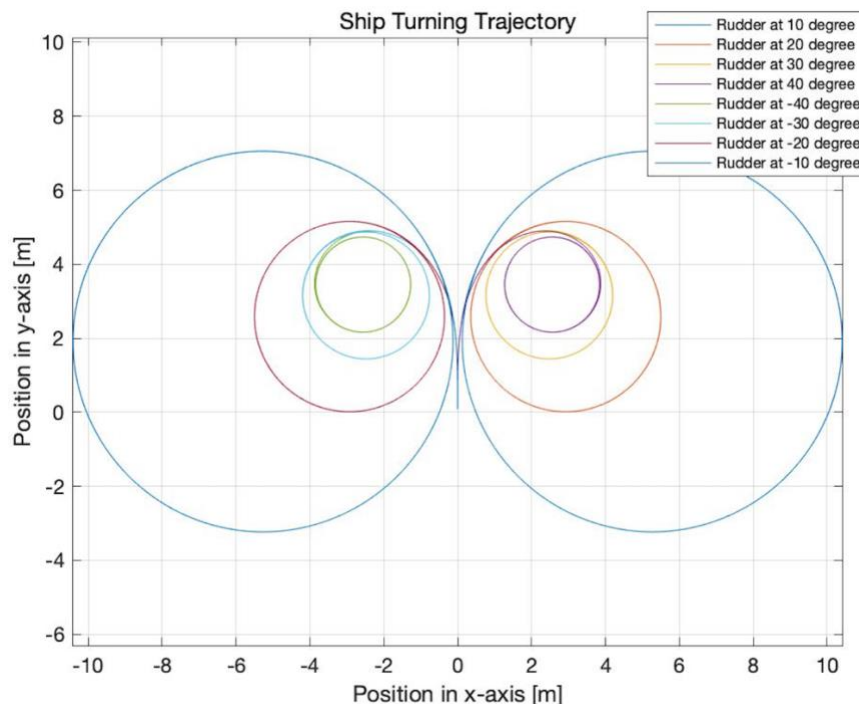


Figure 8: Ship turning trajectory with different angle

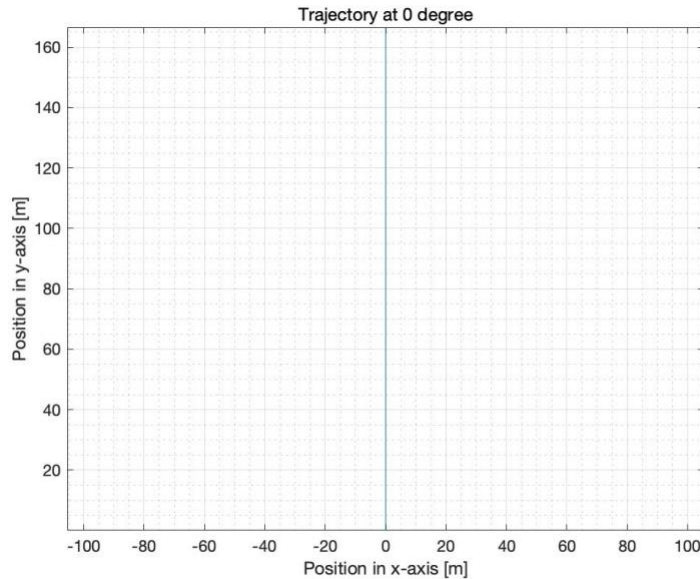


Figure 9: Ship turning trajectory with 0 degree

4 Discussions and Assessment of Results

Compared with the previous researcher's simulation, this simulation result is more accurate. Figure 8 shows the 1/20 Bluefin model turning trajectory with different turning angle by using MMG model. The maximum diameter of the rudder's turning circle of 10, 20, 30, 40 degrees are about 12m, 5.2m; 4.2m and 3.9m respectively. The simulation results show that the larger the rudder angle is, the smaller the diameter is. Figure 9 shows the ship turning trajectory with rudder 0 degree. It can be seen from the simulation that when the rudder angle is 0, the ship's trajectory is a straight line along the y-axis.

5 Conclusions

This paper focuses on the study of the ship's turning motion. MMG model is used in the construction of the ship's mathematical model. In the whole calculation process, because some parameters are obtained by looking up tables and reading graphs, and some parameters can only be estimated based on the parameters of some real ships, the ship maneuverability index obtained is not very accurate. According to the simulation results of MATLAB, we can see that the formula is feasible and shows the scaled Bluefin model can be used for the experiment.

Due to the recent spread of the Covid-19 worldwide, I was unable to return to Australia on time and participate in my experiment. It is necessary to verify the reliability of the mathematical model by comparing the turning radius given by GPS through experiments with the turning radius simulated by MATLAB. Last semester, I participated in the experiments of other researchers, but due to the influence of windy weather, the damage of Bluefin's engine led to the deviation of Bluefin's running trajectory and the loss of computer control during the experiment. In the follow-up experiments, it is recommended to use a GPS base station or replace the GPS sensor to increase the GPS signal to obtain a more accurate trajectory.

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